

NFL Scoring Trends Case Study Rubric

DS 4002 – Spring 2026 – Noah Laidhold

Why am I doing this?

This case study allows you to use data science techniques to analyze how offensive strategy and game pace impact scoring in the NFL. You will work with real-world data to explore trends, build models, and interpret results in a practical context.

What am I going to do?

You will use a dataset derived from nflfastR play-by-play data from the 2010–2024 NFL seasons. The data has been cleaned and aggregated to the weekly level and includes variables such as average points per game, pass rate, and plays per game.

You will conduct exploratory data analysis to examine trends, then build a regression model to evaluate how pass rate and plays per game relate to scoring. Finally, you will interpret your results and identify the strongest predictors of scoring.

Your final deliverables include:

- A Jupyter Notebook with EDA and modeling
- Visualizations showing trends and relationships
- A written summary of findings
- A GitHub repository containing all materials

How will I know I have succeeded?

You will meet expectations on this case study when you successfully follow and complete the criteria in the rubric below:

Spec Category	Spec Details
Formatting	One GitHub repository (submitted via link on Canvas) • Create a repository titled “NFL-Scoring-Case-Study” • Repository must include: ○ README.md ○ LICENSE.md ○ Source Code File(s)

	○ Data
README.md	● Brief summary of the case study ● Explanation of the dataset and materials ● Description of the purpose of the analysis ● Clear explanation of the problem being studied (NFL scoring trends)
Source Code File(s)	Well documented Jupyter Notebook (and/or R script) that includes: ● Data cleaning and preparation steps (if applicable) ● Exploratory data analysis: ○ Trends in scoring over time ○ Trends in pass rate and plays per game ○ Visualizations showing relationships between variables ● Regression modeling: ○ Model using pass rate, plays per game, and time ○ Interpretation of coefficients and statistical significance ● Results: ○ Model performance (R^2 or accuracy) ○ Identification of the strongest predictor of scoring ● Clear comments throughout explaining each step
Data	● Dataset used for analysis must be included in the repository ● Data should be clearly labeled and easy to interpret ● Include cleaned dataset (e.g., weekly aggregated data) ● Any additional datasets used should also be included
REFERENCES.md	● Markdown file titled REFERENCES.md ● Include all sources used in IEEE citation format ● Include brief annotations explaining how each source contributed to the project